Physics Worksheet

Electric Potential

Name: ____________________ **Student #:** _______________ **Student Email #:** ________

1. Electric potential increases along the x axis. A proton and an electron are placed at rest at the origin; in which direction do the charges move when released? Draw a diagram

2. An electron experiences a change in potential of 200 V. What is its speed?

3. Draw a diagram of a capacitor and the field lines of the electric field inside and outside of the capacitor

4. You friend has a capacitor. They want it to have a greater capacitance, so they tell you they are going to pull the two plates of their capacitor farther apart from each other to make more room for charges inside. Is this wise?

5. Consider a capacitor

   (a) The capacitor is filled with an insulating material that has a permittivity twice that of free space. It is comprised of 1.9 m radius circular plates, seperated by 1 μm. What is its capacitance?

   (b) How much power would it take to fill the capacitor with 3 C of charge in 4 ms?